

- Description

This module manages a smart tire pressure monitoring system, ensuring that:

1. **The tire pressure is continuously measured to ensure optimal performance and safety.**
2. **If the tire pressure falls below the minimum threshold, a low-pressure alert is triggered.**
3. **If the tire pressure exceeds the maximum threshold, a high-pressure alert is issued.**
4. **If the tire pressure remains within the optimal range, the system indicates a normal state.**

This system helps maintain vehicle safety and performance by ensuring tire pressure remains within acceptable limits.

- Specification

module SmartTirePressureMonitoring

var currentPressure, minPressure, maxPressure: real

process updatePressure(changeAmount: real)

functional scenario form:

currentPressure < minPressure && output "Low Tire Pressure Alert: " + currentPressure + " psi", Low tire pressure alert is triggered.

||

currentPressure > maxPressure && output "High Tire Pressure Alert: " + currentPressure + " psi", High tire pressure alert is triggered.

||

currentPressure <= maxPressure && output "Tire Pressure is Optimal: " + currentPressure + " psi", Tire pressure is within the optimal range.

end process

end module

- Java Code

```
package code;
class SmartTire {
    private double minPressure;
    private double maxPressure;
    private double currentPressure;
    public SmartTire(double minPressure, double maxPressure, double initialPressure) {
        this.minPressure = minPressure;
        this.maxPressure = maxPressure;
        this.currentPressure = initialPressure;
    }
    public void updatePressure(double changeAmount) {
        currentPressure += changeAmount;
    }
}
```

```

        System.out.printf("Current Tire Pressure: %.2f psi\n", currentP
ressure);
        if (currentPressure < minPressure) {
            System.out.println("ALERT: Low Tire Pressure!");
        } else if (currentPressure > maxPressure) {
            System.out.println("ALERT: High Tire Pressure!");
        } else {
            System.out.println("Tire Pressure is Optimal.");
        }
    }
}

public class c69_SmartTireSystem {
    public static void main(String[] args) {
        SmartTire tire = new SmartTire(30.0, 35.0, 33.0);

        // testcase-VT:
        //     tire.updatePressure(-5.0);
        //     tire.updatePressure(8.0);
        //     tire.updatePressure(-2.0);

        // testcase-FT:
        //     tire.updatePressure(8.812221249325049E307);
        //     tire.updatePressure(7.986142550757732E295);
        //     tire.updatePressure(3.4300223128453513E305);

        // testcase-Z3:
        //     tire.updatePressure(-5.0);
        //     tire.updatePressure(8.0);
        //     tire.updatePressure(-2.0);

        // testcase-UVT:
        tire.updatePressure(-5.0);
        tire.updatePressure(8.0);
        tire.updatePressure(-2.0);
    }
}

```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	838	842
▼ src	0.5 %	4	838	842
▼ code	0.5 %	4	838	842
▼ c69_SmartTireSystem.java	100.0 %	4	0	4
> c69_SmartTireSystem		0	0	0
> SmartTire	100.0 %	4	0	4

3. Vibration testing instruction coverage: 95.9%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.3 %	71	20,805	20,876
✓ src	0.3 %	71	20,805	20,876
✓ code	0.3 %	71	20,805	20,876
✓ c69_SmartTireSystem.java	95.9 %	71	3	74
> c69_SmartTireSystem	85.0 %	17	3	20
> SmartTire	100.0 %	54	0	54

4. Fuzz testing branch coverage: 50%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.2 %	2	840	842
✓ src	0.2 %	2	840	842
✓ code	0.2 %	2	840	842
✓ c69_SmartTireSystem.java	50.0 %	2	2	4
> c69_SmartTireSystem		0	0	0
> SmartTire	50.0 %	2	2	4

5. Fuzz testing instruction coverage: 86.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.3 %	64	20,812	20,876
✓ src	0.3 %	64	20,812	20,876
✓ code	0.3 %	64	20,812	20,876
✓ c69_SmartTireSystem.java	86.5 %	64	10	74
> c69_SmartTireSystem	85.0 %	17	3	20
> SmartTire	87.0 %	47	7	54

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.5 %	4	838	842
✓ src	0.5 %	4	838	842
✓ code	0.5 %	4	838	842
✓ c69_SmartTireSystem.java	100.0 %	4	0	4
> c69_SmartTireSystem		0	0	0
> SmartTire	100.0 %	4	0	4

7. Z3 instruction coverage: 95.9%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.3 %	71	20,805	20,876
✓ src	0.3 %	71	20,805	20,876
✓ code	0.3 %	71	20,805	20,876
✓ c69_SmartTireSystem.java	95.9 %	71	3	74
> c69_SmartTireSystem	85.0 %	17	3	20
> SmartTire	100.0 %	54	0	54

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.5 %	4	838	842
✓ src	0.5 %	4	838	842
✓ code	0.5 %	4	838	842
✓ c69_SmartTireSystem.java	100.0 %	4	0	4
> c69_SmartTireSystem		0	0	0
> SmartTire	100.0 %	4	0	4

9. Unimproved V-method instruction coverage: 95.9%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.3 %	71	20,805	20,876
✓ src	0.3 %	71	20,805	20,876
✓ code	0.3 %	71	20,805	20,876
✓ c69_SmartTireSystem.java	95.9 %	71	3	74
> c69_SmartTireSystem	85.0 %	17	3	20
> SmartTire	100.0 %	54	0	54